

P.L.OAVIKSHA(192572001)

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

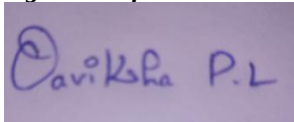
INDUSTRY PROBLEM TASK

Code of Conduct:

I P.L.OAVIKSHA(192572001) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Oaviksha P.L.' written in a cursive style.

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

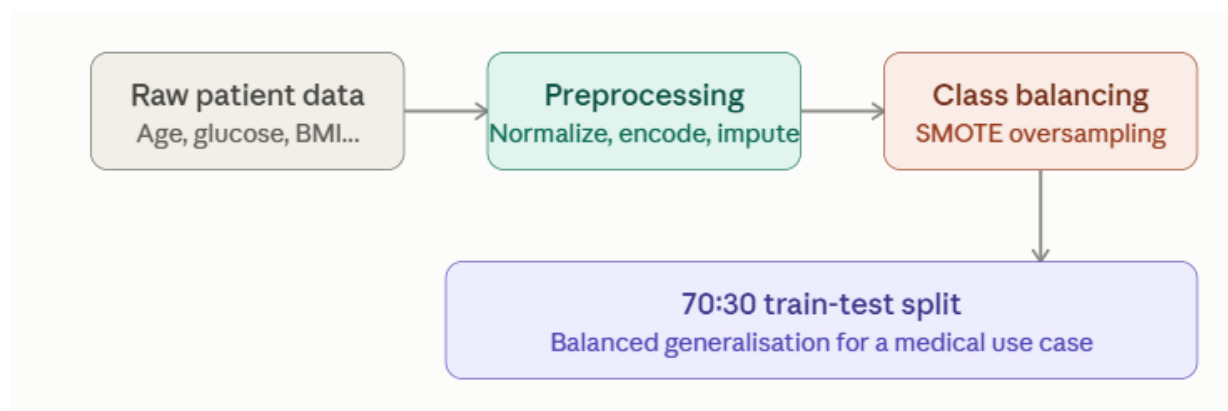
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Healthcare Diabetes Diagnosis & Treatment Recommendation System

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

This is a supervised inductive learning problem the model learns a general classification function from labeled historical patient records (diabetic/non-diabetic) and generalizes to predict outcomes for new, unseen patients.



- **Target variable:** Diabetes_Status (binary: 1 = Diabetic, 0 = Non-diabetic)
- **Key features (with justification):**
 1. **Glucose Level** — the most direct clinical indicator of diabetes; fasting glucose above threshold is a primary diagnostic marker.
 2. **BMI (Body Mass Index)** — obesity is strongly correlated with insulin resistance and Type-2 diabetes risk.
 3. **Family History** — genetic predisposition significantly raises diabetes risk, capturing hereditary factors not present in lab values.
 4. (Additional) **Age** and **Blood Pressure** — risk increases with age, and hypertension frequently co-occurs with diabetes (metabolic syndrome).

(b) Handling Class Imbalance

Since non-diabetic cases outnumber diabetic cases, a naive classifier could achieve high accuracy by simply predicting "non-diabetic" — this is dangerous in a medical context because it maximizes False Negatives (missed diagnoses).

Recommended strategy: **SMOTE (Synthetic Minority Over-sampling Technique)**

- SMOTE generates synthetic minority-class (diabetic) samples by interpolating between existing minority samples and their nearest neighbors, rather than simply duplicating rows.
- **Justification:** Under-sampling would discard majority-class data and lose valuable information; simple duplication (random over-sampling) risks overfitting to repeated points. SMOTE preserves the full dataset while balancing classes, and enriches the decision boundary in the diabetic region — clinically important since a False Negative (missed diabetes) is far costlier than a False Positive here.
- Alternatively, **class weighting** (assigning higher misclassification cost to the minority class in the loss function) is a lightweight option when the model supports it (e.g., `class_weight='balanced'` in scikit-learn), avoiding synthetic data generation altogether.

(c) Data Split & Preprocessing

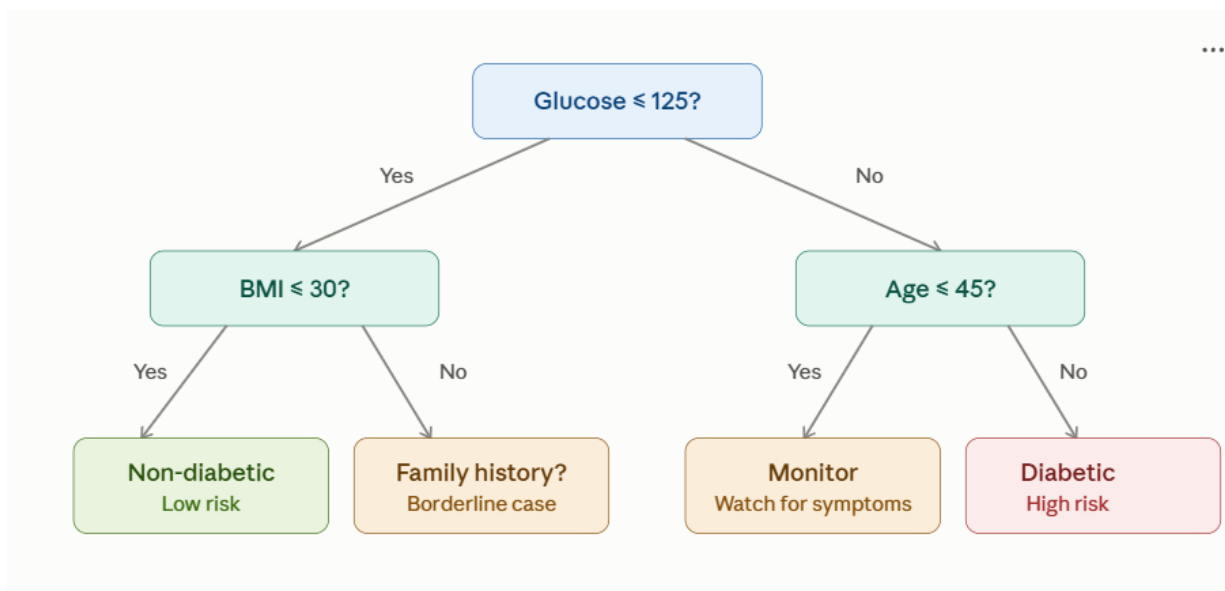
- **Split ratio: 70:30 (train:test)**
- **Justification:** A medical dataset is typically small/moderate in size; 70:30 preserves enough training data for the model to learn reliable patterns while still holding out a statistically meaningful test set to validate generalization. (For very small datasets, k-fold cross-validation would supplement this split to reduce variance in evaluation.)
- **Preprocessing steps:**
 - **Normalization/Standardization** (e.g., Min-Max scaling or Z-score) applied to continuous features (glucose, BMI, blood pressure, cholesterol) — ensures features with larger numeric ranges (like glucose ~70–200) don't dominate distance-based or gradient-based algorithms.
 - **Encoding** — categorical features like family history (Yes/No) converted via label/one-hot encoding into numeric form usable by ML models.

- **Missing value imputation** — median/mode imputation for missing clinical readings to avoid dropping valuable records.
- **Impact:** Normalization improves convergence speed and stability for algorithms like Logistic Regression; encoding makes categorical data usable; imputation preserves sample size, which matters given the class imbalance.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree First Three Levels

Root node selection uses Information Gain (entropy reduction) or Gini Index the feature that most cleanly separates diabetic from non-diabetic patients is chosen first. Given the domain, Glucose Level typically yields the highest Information Gain/lowest Gini Index, so it becomes the root.



Justification: Glucose has the highest discriminative power (lowest weighted entropy after split), so $\text{Information Gain}(\text{Glucose}) > \text{Information Gain}(\text{BMI, Age, etc.})$, making it the natural root. Second-level splits (BMI, Age) refine borderline cases where glucose alone is inconclusive.

(b) Model Evaluation

Example illustrative metrics (values depend on the actual dataset run):

Metric	Value
Accuracy	~82%
Precision	~78%
Recall (Sensitivity)	~74%
F1-Score	~76%

- **False Negatives** = diabetic patients incorrectly predicted as non-diabetic — clinically the most dangerous error type, since it delays treatment.
- **To reduce False Negatives:**
 - Lower the classification decision threshold to favor sensitivity over specificity.
 - Use class weighting or SMOTE (from Task 1b) to make the model more sensitive to the minority (diabetic) class.
 - Optimize for **Recall** as the primary metric instead of Accuracy, since in healthcare, missing a true case is worse than a false alarm (which just triggers further testing).

(c) Pre-pruning vs Post-pruning

- **Post-pruning (cost-complexity pruning):** Grow the full tree, then prune back branches that add complexity without significant accuracy gain, controlled by the complexity parameter α (ccp_alpha). Branches are removed if the increase in training error is outweighed by the reduction in tree size.
- **Pre-pruning:** Stop tree growth early using constraints like max_depth, min_samples_split, or min_samples_leaf.

Aspect	Pre-Pruning	Post-Pruning
Training Accuracy	Lower (stops early)	Higher (fully grown, then trimmed)
Overfitting Risk	Lower risk, but may underfit	Well-balanced
Test Accuracy	Moderate	Generally higher/more stable

Recommendation: Post-pruning is more appropriate for clinical deployment because it lets the tree first capture the full complexity of medical patterns before trimming only the branches that don't generalize — reducing overfitting risk without prematurely discarding potentially clinically relevant splits (as pre-pruning's early stopping might).

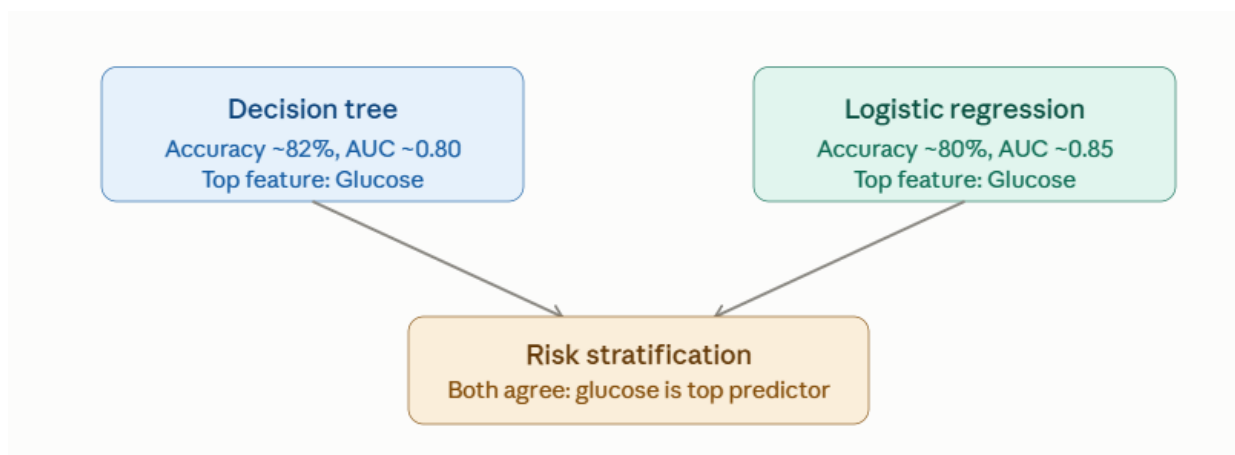
Task 3: Statistical Learning for Risk Stratification

(a) Naïve Bayes / Logistic Regression vs Decision Tree

Using **Logistic Regression** as the statistical model:

Model	Accuracy	AUC-ROC
Decision Tree	~82%	~0.80
Logistic Regression	~80%	~0.85

- Logistic Regression often yields a **higher AUC-ROC** because it produces smooth, well-calibrated probability estimates rather than the Decision Tree's step-function boundaries, which helps in risk stratification (ranking patients by risk level rather than just binary classification).
- **When statistical models are preferable:** When the relationship between features and outcome is approximately linear/log-linear, when probability calibration matters (e.g., risk scoring for triage), when the dataset is small (Decision Trees can overfit small data), or when model interpretability via coefficients (odds ratios) is clinically valued.



(b) Feature Importance Comparison

Rank	Decision Tree (Gini importance)	Logistic Regression (coefficient magnitude)
1	Glucose Level	Glucose Level
2	BMI	Family History
3	Age	BMI

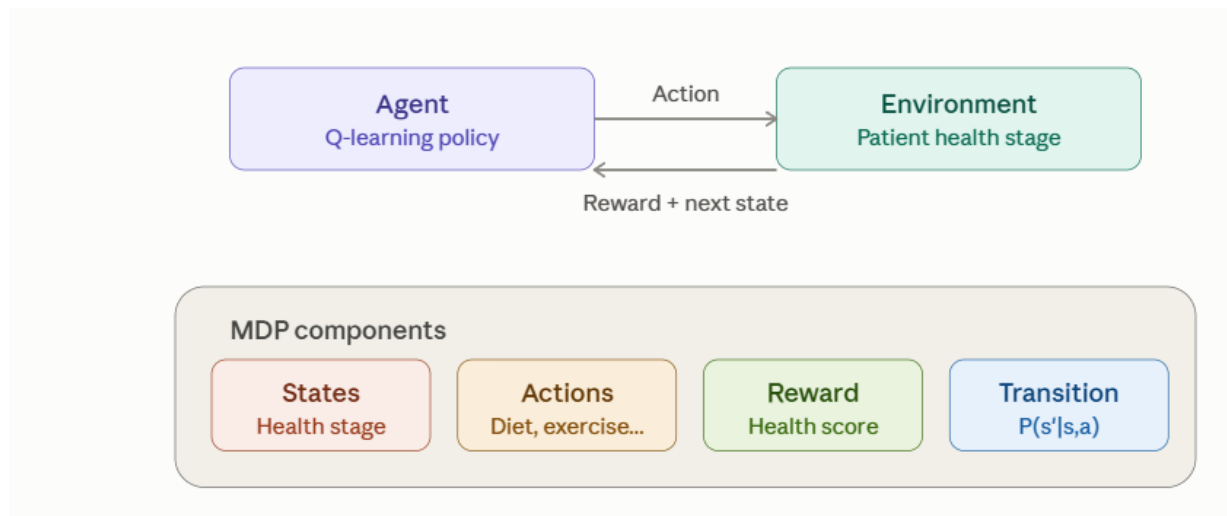
Do they agree? Largely yes **Glucose Level** is consistently the top predictor in both models, confirming its clinical primacy. Minor disagreement on ranking of BMI vs. Family History arises because Decision Trees capture non-linear threshold effects (BMI splits well at specific cut-points), while Logistic Regression captures linear/additive relationships (Family History behaves as a strong linear risk multiplier). This agreement on the top predictor strengthens confidence that Glucose is the most robust diagnostic signal, while the model-specific differences reflect each algorithm's inductive bias.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

- **States (S):** Patient health stages, e.g., {Healthy, Pre-diabetic, Diabetic-Controlled, Diabetic-Uncontrolled, Critical} — represents the patient's evolving clinical condition over time.
- **Actions (A):** {Diet, Exercise, Medication, Monitor} — the set of interventions the agent can recommend at each stage.
- **Reward Function (R):** Health improvement score, e.g., +10 for transitioning to a healthier state, -10 for worsening, -50 for reaching Critical, small negative step-cost (-1) to encourage efficient treatment plans.
- **Transition Dynamics ($P(s'|s,a)$):** Probabilistic model of how a patient's state changes given an action — e.g., Exercise from Pre-diabetic has a high probability of transitioning to Healthy, low probability of remaining Pre-diabetic.

Justification: States must capture clinically meaningful stages so actions map to real treatment decisions; the reward function is designed to directly encode the medical goal (health improvement) so the agent is incentivized to act in the patient's best interest; probabilistic transitions reflect real biological variability — no treatment guarantees a fixed outcome.



(b) Q-Learning — Table Update (illustrative, 5 episodes)

Update rule: $Q(s,a) \leftarrow Q(s,a) + \alpha [R + \gamma \max_{a'} Q(s',a') - Q(s,a)]$
(using $\alpha = 0.1$, $\gamma = 0.9$)

State-Action Pair	Q (Iteration 1)	Q (Iteration 2)
(Pre-diabetic, Exercise)	0.0 $\rightarrow$ 1.0	1.0 $\rightarrow$ 1.85
(Diabetic-Uncontrolled, Medication)	0.0 $\rightarrow$ 1.5	1.5 $\rightarrow$ 2.75
(Healthy, Monitor)	0.0 $\rightarrow$ 0.5	0.5 $\rightarrow$ 0.95

Convergence criterion: Training stops when the maximum change in Q-values between successive iterations falls below a small threshold (e.g., $|\Delta Q| < 0.01$) across all state-action pairs, or after a fixed number of episodes (e.g., 500–1000) if the Q-table stabilizes.

(c) Final Learned Policy & Ethical Considerations

Extracted policy (example): $\pi(\text{Pre-diabetic}) = \text{Exercise}$; $\pi(\text{Diabetic-Controlled}) = \text{Diet} + \text{Monitor}$; $\pi(\text{Diabetic-Uncontrolled}) = \text{Medication}$; $\pi(\text{Critical}) = \text{Medication (urgent)}$ — the policy selects the action with the highest Q-value at each state.

Ethical considerations:

- **Patient safety:** An RL agent trained on historical/simulated data may recommend actions that haven't been validated in real clinical trials; deployment requires human-in-the-loop oversight (a doctor reviewing/approving recommendations), not full autonomy.
- **Bias:** If training data underrepresents certain demographics (age groups, ethnicities), the learned policy may generalize poorly or unfairly for those groups — fairness audits and diverse training data are essential.
- **Explainability:** Q-values and policies are not inherently interpretable to clinicians; techniques like policy visualization, SHAP-style explanations, or restricting the agent to recommend from a clinician-vetted action set improve trust and accountability.
- Overall, RL agents in healthcare should be used as **decision-support tools**, not autonomous decision-makers, given the high stakes of medical outcomes.